

Decision-Aware Physics-Informed Forecasting and Metaheuristic Allocation for NHS Critical-Care Surge Capacity Under Demand Uncertainty

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Abstract. Epidemic waves create rapid, spatially uneven pressure on critical-care services, requiring health systems to decide where to expand surge capacity and how to allocate patients under demand uncertainty. We propose a forecast-to-decision optimisation framework for critical-care surge capacity planning in NHS England. A regional physics-informed neural epidemic forecasting module estimates multi-horizon demand using a refined $SEI_a I_s HCRD$ compartmental model with distinct hospitalisation and critical-care escalation parameters, trained with a decision-aware loss that asymmetrically penalises forecast underestimation. Quantile forecasts are discretised into demand scenarios that drive a robust multi-objective location-allocation model balancing infrastructure cost, unmet demand, patient-transfer burden, and equity, subject to capacity, budget, and travel-time constraints. We compare exact mixed-integer linear programming, scenario-based robust MILP, a Genetic Algorithm with linear-programming slave, the Non-dominated Sorting Genetic Algorithm II, and Simulated Annealing on NHS England COVID-19 hospital activity and critical-care capacity data. Allocation policies are evaluated on unmet demand, coverage, mean travel burden, worst-case shortage, and a regional shortage-equity index. The framework links predictive modelling with prescriptive optimisation to support transparent, robust, and equitable critical-care surge planning.

Keywords: physics-informed neural networks, robust optimisation, metaheuristics, NSGA-II, critical-care planning, epidemic forecasting, NHS England

1 Introduction

The COVID-19 pandemic exposed the vulnerability of national health systems to rapid, spatially uneven surges in critical-care demand. In England, sustained demand for adult critical-care beds and mechanical ventilation during successive waves required emergency capacity expansion, inter-trust patient transfers, and Nightingale-style temporary hospitals. Decisions of this scale are made under deep uncertainty: planners must balance the cost of opening or expanding surge

capacity against the risk of unmet demand, while remaining sensitive to patient-transfer burden and regional equity. The computational intelligence community has produced strong tools for both halves of this problem — epidemic and hospital-demand forecasting [10,9,16] and robust facility-location and resource allocation under uncertainty [8,20,17] — but few works connect the two into a single decision-focused pipeline tailored to a national health system. The present work occupies the *temporal, per-region* corner of the forecasting design space and the *multi-method comparative* corner of the allocation design space; the spatiotemporal-graph alternatives surveyed in Section 2 are explicitly deferred to future work.

Existing forecasting models, including physics-informed neural networks (PINNs) [1,16] and graph-based architectures such as STAN [10] and HOIST [9], are typically optimised for statistical accuracy alone. Yet for critical-care planning the operational cost of forecast errors is asymmetric: underestimating demand by ten beds risks patients without ventilators, while overestimating wastes prepared capacity. Standard squared-error or absolute-error objectives treat these symmetrically. Conversely, most optimisation formulations for pandemic capacity planning assume demand is deterministic or drawn from off-the-shelf statistical forecasts [17,8], leaving unexploited the dynamics-aware information that physics-informed forecasters can provide. Recent work begins to close this gap: Liu and Cao [14] couple spatiotemporal graph attention with multi-objective optimisation for vaccine and non-pharmaceutical-intervention allocation, and Shams Eddin and El Hajj [17] integrate SEIRD-driven demand into robust hospital-bed scheduling for a single US region. To our knowledge no published work combines (a) physics-informed regional epidemic forecasting, (b) decision-aware training that explicitly penalises operationally costly forecast errors, and (c) metaheuristic robust optimisation for NHS England critical-care surge capacity.

We close that gap with a four-phase forecast-to-decision pipeline. A per-region PINN-SEIRD module produces multi-horizon point and quantile forecasts of admissions and critical-care occupancy, trained with a composite loss combining a Huber data fit, an ODE-residual physics regulariser, an asymmetric underestimation penalty, and a temporal smoothness term. Forecast quantiles are discretised into demand scenarios that drive a robust multi-objective location-allocation model, solved exactly via mixed-integer linear programming (MILP) at NHS regional scale, and via metaheuristics — a Genetic Algorithm (GA), the Non-dominated Sorting Genetic Algorithm II (NSGA-II) [6], and Simulated Annealing — at trust-level scale. Our contributions are: (1) a four-phase forecast-to-decision pipeline reproducible on open NHS data; (2) a *decision-aware* training objective penalising operationally costly forecast errors; (3) a robust multi-objective allocation model balancing infrastructure cost, unmet demand, transfer burden, and equity, with explicit MILP/GA/NSGA-II/SA comparison; (4) a refined $SEI_a I_s HCRD$ compartmental model that separates the hospitalisation ratio ω ($I_s \rightarrow H$) from the critical-care escalation proportion ϕ ($H \rightarrow C$), correcting an over-loading of ω in earlier work [1]; and (5) an NHS England case study evaluating allocation policies against operational, equity, and robustness metrics.

Section 2 reviews related work; Sections 3 and 4 describe the forecasting and optimisation modules; Section 5 presents the case study; Sections 6 to 8 report results, discuss limitations, and conclude.

2 Related Work

The closest precedent for our pipeline is Bertsimas et al. [3], who couple an extended DELPHI compartmental model of COVID-19 to a bilinear mixed-integer program for the location of one hundred mass-vaccination sites across US states, solving the resulting non-convex problem by a coordinate-descent scheme that alternates forward simulation of the discretised dynamics with a linear-programming approximation of the bilinear constraints. Their pipeline is the methodological parallel to ours; the differences are domain (vaccination-site siting versus critical-care surge), coupling (iterative joint optimisation versus our decoupled forecast-then-optimize), and forecasting objective (mean-squared error on DELPHI states versus our decision-aware composite loss). Our framework also draws on the formal Smart Predict-then-Optimize framework of Elmachet and Grigas [7] and on the neural-ODE-based equitable opioid facility-location work of Luo and Stellato [15], who fit per-state SIR-like models via a neural-ODE-inspired procedure and embed the discretised dynamics into a McCormick-relaxed MIP. We review the three relevant subliteratures below.

2.1 Epidemic and Hospital-Demand Forecasting

Statistical and recurrent baselines—ARIMA, Prophet, and LSTM/GRU architectures—remain competitive when applied per region [2]. Closest in target to our work, Valente et al. [19] forecast COVID-19 intensive-care occupancy in the Lazio region of Italy using a bidirectional LSTM with Google Community Mobility features lagged by twenty-one days following Cartení et al. [5], and exploit stochastic-ensemble Monte-Carlo dropout for prediction intervals; they report mean absolute error of approximately seven beds on weekly horizons. Graph neural networks add explicit spatial coupling; STAN [10] combines graph-attention pooling with an SIR-consistency loss for COVID-19 case prediction at US county and state level, and HOIST [9] introduces an Ising-dynamics-inspired regulariser over an adaptively-learned location graph for hospitalisation prediction across 2,299 US counties. Physics-informed neural networks embed compartmental ordinary differential equations directly into the training loss [1,16], supporting dynamics-consistent forecasts under regime shifts. NHS-specific work is sparse: Hunter et al. [11] target elective-surgery ICU demand at a single trust, and prior work by the present authors [2,1] forecasts national-scale mechanical-ventilation demand. None of these targets the multi-region NHS England critical-care planning problem, and all optimise statistical rather than operationally-aligned error metrics.

2.2 Robust Optimisation for Healthcare Resource Allocation

Pandemic capacity planning has produced a rich optimisation literature. Eriskin et al. [8] develop a robust multi-objective MILP for emergency-facility location and patient assignment under demand uncertainty. Wan et al. [20] propose a two-stage multi-objective ICU bed allocation model solved by an improved NSGA-II on Lyon hospital data. Shams Eddin and El Hajj [17] integrate SEIRD-driven demand with standard beds, buffer beds, field hospitals, and inter-hospital transfers, validated on Northern Virginia data. In the vaccine supply-chain space, Kargar et al. [12] couple a Covasim agent-based simulator [13] to a multi-echelon bi-objective MILP solved by the augmented ε -constraint method, exposing the cost-versus-emissions Pareto front; we adopt the augmented ε -constraint paradigm in our allocation comparison, complementing exact MILP and NSGA-II with an exact non-dominated-solution generator. Tang et al. [18] apply adaptive large neighbourhood search to mobile-vaccination scheduling, exemplifying the broader operational-research-meets-public-health literature beyond static facility location. Metaheuristic methods—GA, NSGA-II [6], simulated annealing, and ALNS—are routinely employed when problem scale defeats exact MILP. The bulk of this literature is single-region or US-focused; trust-level allocation across NHS England regions and Integrated Care Boards has, to our knowledge, not been studied with the methodological breadth proposed here.

2.3 Forecast-to-Decision Pipelines

The integration of predictive forecasts into prescriptive optimisation is a maturing research area. Elmachetoub and Grigas [7] formalise the Smart Predict-then-Optimize framework with a loss aware of the downstream decision problem. In healthcare, Bertsimas et al. [3] provide the closest end-to-end precedent via coordinate-descent coupling of DELPHI simulation with bilinear MIP allocation; Luo and Stellato [15] provide an analogous predict-then-optimize architecture for opioid treatment-facility location with neural-ODE parameter estimation and McCormick-envelope relaxation; and Liu and Cao [14] couple spatiotemporal graph-attention forecasting to multi-objective vaccine and non-pharmaceutical-intervention allocation. Our framework adopts the predict-then-optimize structure, contributes a domain-specific decision-aware training objective that asymmetrically penalises operationally costly under-prediction, and broadens the optimisation comparison to include exact MILP, augmented ε -constraint, NSGA-II, GA, and simulated annealing. By treating the forecasting module as a substitutable component, the framework is forecasting-backend agnostic: alternative graph-based or transformer-based forecasters can be swapped in without changes to the allocation layer.

3 Forecasting Module

The forecasting module produces multi-horizon point and quantile estimates of critical-care demand for each NHS region. It comprises three parts: a per-region

physics-informed neural epidemic estimator, a temporal forecasting head, and a Monte Carlo Dropout uncertainty layer.

3.1 Refined SEIRD Compartmental Model

For each region $r \in \mathcal{R}$ with population N_r , we model the local epidemic using a refined eight-compartment $SEI_a I_s HCRD$ system. Let $U^r(t) = (S_r, E_r, I_{a,r}, I_{s,r}, H_r, C_r, R_r, D_r)$ denote susceptible, exposed, asymptomatic infectious, symptomatic infectious, hospitalised, critical-care, recovered, and deceased compartments respectively. The system evolves according to

$$\begin{aligned}
 \dot{S}_r &= -\beta_r S_r (I_{s,r} + I_{a,r}) / N_r + \eta_r R_r \\
 \dot{E}_r &= \beta_r S_r (I_{s,r} + I_{a,r}) / N_r - \alpha E_r \\
 \dot{I}_{s,r} &= \alpha \rho E_r - d_s I_{s,r} \\
 \dot{I}_{a,r} &= \alpha (1 - \rho) E_r - d_a I_{a,r} \\
 \dot{H}_r &= d_s \omega I_{s,r} - d_H H_r - \mu H_r \\
 \dot{C}_r &= \phi d_H H_r - \gamma_{c,r} C_r - \delta_{c,r} C_r \\
 \dot{R}_r &= d_s (1 - \omega) I_{s,r} + d_a I_{a,r} + (1 - \phi) d_H H_r + \gamma_{c,r} C_r - \eta_r R_r \\
 \dot{D}_r &= \mu H_r + \delta_{c,r} C_r,
 \end{aligned} \tag{1}$$

subject to conservation $\sum_{\bullet} \bullet_r(t) = N_r$. Here α, d_s, d_a, d_H are *rates* with units of inverse-days (their reciprocals are the corresponding clinical periods), ρ is the symptomatic proportion, ω is the hospitalisation ratio of symptomatic cases, μ is the in-hospital mortality rate, and $\beta_r, \gamma_{c,r}, \delta_{c,r}, \eta_r$ are time-varying region-specific parameters learned from data.

Refinement of the prior compartmental model. The earlier formulation in [1] overloaded the parameter ω to represent both the $I_s \rightarrow H$ hospitalisation ratio and, via $(1 - \omega)$, the $H \rightarrow C$ critical-care escalation. These quantities are biologically distinct: the first is determined by clinical severity at admission, the second by in-hospital deterioration. We separate them by introducing ϕ for the $H \rightarrow C$ escalation, leaving ω for hospitalisation alone. Equation (1) reflects this refinement.

3.2 Per-Region PINN-SEIRD

For each region r , the PINN consists of two fully connected feedforward networks following [1]: a state network $U_{NN}^r : t \mapsto U^r(t)$ with five hidden layers of 20 units, and a parameter network $X_{NN}^r : t \mapsto (\beta_r, \gamma_{c,r}, \delta_{c,r}, \eta_r)$ with three hidden layers of 20 units. Both use tanh activations, sigmoid outputs, and Xavier initialisation. The PINN loss for region r combines a data-fit term on observable compartments

$\{I_s, H, C, D\}$ with a residual loss against Equation (1):

$$\mathcal{L}_r^{PINN} = \sum_{t \in \mathcal{T}_{\text{train}}} \sum_{k \in \{I_s, H, C, D\}} (U_{NN,k}^r(t) - y_{r,t,k})^2 + \lambda_{ode} \sum_{t \in \mathcal{T}_{\text{coll}}} \left\| \frac{dU_{NN}^r(t)}{dt} - \mathcal{N}(U_{NN}^r(t); X_{NN}^r(t)) \right\|^2, \quad (2)$$

where $\mathcal{N}$ denotes the right-hand side of Equation (1) and derivatives are evaluated via PyTorch automatic differentiation at random collocation points sampled in $\mathcal{T}_{\text{coll}}$ at every iteration.

3.3 Temporal Forecasting Head and Decision-Aware Loss

Once the PINN is trained, its outputs — region-specific compartmental state estimates $\tilde{U}^r(t)$ and time-varying parameters $\tilde{X}^r(t)$ — are concatenated with observed covariates $\mathbf{x}_{r,t}$ to form the feature sequence $\mathbf{f}_{r,t} = [\tilde{U}^r(t) \parallel \tilde{X}^r(t) \parallel \mathbf{x}_{r,t}]$. A two-layer GRU with hidden dimension 64 encodes a $L = 28$ day lookback window into a representation $\mathbf{h}_{r,t}$, from which a multi-horizon decoder produces $\hat{y}_{r,t+h} = \mathbf{w}_h^\top \mathbf{h}_{r,t} + b_h$ for $h \in \mathcal{H} = \{7, 14, 21, 28\}$.

The forecaster is trained with a composite *decision-aware* loss:

$$\mathcal{L} = \mathcal{L}_{\text{forecast}} + \lambda_{\text{phys}} \sum_r \mathcal{L}_r^{PINN} + \lambda_{\text{under}} \mathcal{L}_{\text{under}} + \lambda_{\text{smooth}} \mathcal{L}_{\text{smooth}}, \quad (3)$$

where $\mathcal{L}_{\text{forecast}}$ is the Huber loss on standardised forecasts (robust to data spikes), and the remaining terms are

$$\mathcal{L}_{\text{under}} = \sum_{r,h} \max(0, y_{r,t+h} - \hat{y}_{r,t+h}), \quad (4)$$

$$\mathcal{L}_{\text{smooth}} = \sum_{r,h} (\hat{y}_{r,t+h+1} - \hat{y}_{r,t+h})^2. \quad (5)$$

The asymmetric underestimation term $\mathcal{L}_{\text{under}}$ is the methodological hook that ties forecasting to optimisation: it penalises forecast errors only in the direction that creates downstream operational shortages [7]. Initial loss weights are $\lambda_{\text{phys}} = 0.1$, $\lambda_{\text{under}} = 0.5$, $\lambda_{\text{smooth}} = 0.01$, tuned on the validation split.

3.4 Uncertainty and Scenario Generation

Predictive uncertainty is obtained via Monte Carlo Dropout: at inference, dropout layers remain active and $K = 100$ stochastic forward passes generate samples from which empirical quantiles $q_{r,h}^p$ for $p \in \{0.1, 0.5, 0.9\}$ are extracted. Each region-horizon predictive distribution is discretised into three canonical scenarios: $\mathcal{S} = \{(q_{r,h}^{0.1}, 0.2), (q_{r,h}^{0.5}, 0.6), (q_{r,h}^{0.9}, 0.2)\}$. Calibration is reported using the Weighted Interval Score [4] at the 50% and 90% prediction intervals.

4 Robust Allocation Model

The allocation module decides which surge facilities to open or activate, how much additional critical-care capacity to provide at each, and how patient demand should be assigned across NHS regions, given forecast scenarios from Section 3. We present a deterministic mixed-integer linear programme, a scenario-based robust extension, and three metaheuristic solution methods.

4.1 Sets, Parameters and Decision Variables

Let $\mathcal{I}$ index demand nodes (NHS regions for the regional formulation; NHS trusts for the trust-level formulation), $\mathcal{J}$ candidate surge-capacity sites, $\mathcal{H}$ planning horizons, and $\mathcal{S}$ demand scenarios with probabilities π_s . Parameters are: forecast demand $d_{i,h}^s$, baseline capacity $C_{j,h}$, fixed activation cost F_j , marginal capacity cost g_j , per-patient transfer cost c_{ij} , travel time T_{ij} , travel-time threshold τ , total budget B , maximum surge expansion K_j , and population p_i . Decision variables are: the binary opening/activation x_j , additional capacity $b_{j,h} \geq 0$, demand assignment $z_{ij,h}^s \geq 0$, and unmet demand $u_{i,h}^s \geq 0$. Auxiliary variables θ_i, θ linearise the equity penalty and W tracks worst-case shortage in the robust extension.

4.2 Deterministic MILP Formulation

The deterministic model minimises a weighted sum of infrastructure cost, patient-transfer cost, expected unmet demand, and a regional equity penalty:

$$\min \underbrace{\sum_j F_j x_j + \sum_{j,h} g_j b_{j,h}}_{\text{infrastructure}} + \underbrace{\sum_{i,j,h,s} \pi_s c_{ij} z_{ij,h}^s}_{\text{transfer}} + \lambda_1 \underbrace{\sum_{i,h,s} \pi_s u_{i,h}^s}_{\text{shortage}} + \lambda_2 \theta, \quad (6)$$

subject to demand satisfaction, capacity, expansion, budget, travel-time and equity constraints:

$$\sum_j z_{ij,h}^s + u_{i,h}^s \geq d_{i,h}^s \quad \forall i \in \mathcal{I}, h \in \mathcal{H}, s \in \mathcal{S}, \quad (7)$$

$$\sum_i z_{ij,h}^s \leq C_{j,h} + b_{j,h} \quad \forall j \in \mathcal{J}, h \in \mathcal{H}, s \in \mathcal{S}, \quad (8)$$

$$b_{j,h} \leq K_j x_j \quad \forall j \in \mathcal{J}, h \in \mathcal{H}, \quad (9)$$

$$\sum_j F_j x_j + \sum_{j,h} g_j b_{j,h} \leq B, \quad (10)$$

$$z_{ij,h}^s = 0 \quad \text{whenever } T_{ij} > \tau, \quad (11)$$

$$\theta_i \cdot \left(\sum_{h,s} \pi_s d_{i,h}^s / p_i + \varepsilon \right) \geq \sum_{h,s} \pi_s u_{i,h}^s / p_i \quad \forall i \in \mathcal{I}, \quad (12)$$

$$\theta \geq \theta_i \quad \forall i \in \mathcal{I}, \quad (13)$$

$$x_j \in \{0, 1\}, b_{j,h}, z_{ij,h}^s, u_{i,h}^s, \theta_i, \theta \geq 0. \quad (14)$$

Equation (7) ensures that assigned demand plus unmet demand covers forecast demand. Equation (8) caps assignments at total available capacity. The expansion variable $b_{j,h}$ is gated by x_j via Equation (9), so capacity additions can only occur at activated facilities. Equation (11) enforces a maximum travel time τ for any patient transfer. Equations (12) and (13) linearise the *maximum regional shortage ratio per capita* via auxiliaries θ_i and θ , with $\varepsilon = 1$ avoiding division by zero on regions with negligible forecast demand.

4.3 Robust Extension

To penalise tail outcomes we add a worst-case shortage term in CVaR-flavoured form. Define an upper-quantile scenario subset $\mathcal{S}_w \subseteq \mathcal{S}$ (e.g., the High and tail scenarios) and an auxiliary $W \geq \sum_{i,h} u_{i,h}^s$ for all $s \in \mathcal{S}_w$, adding $\lambda_3 W$ to Equation (6). Here $\lambda_3 = 0$ recovers the expected-value formulation and $\lambda_3 \rightarrow \infty$ approaches a worst-case (minimax) policy.

4.4 Solution Methods

Exact MILP (regional scale). For the regional formulation with $|\mathcal{I}| = 7$, $|\mathcal{J}| \leq 15$, $|\mathcal{H}| = 4$, $|\mathcal{S}| = 3$, the model has $O(10^4)$ variables. We solve it exactly using Gurobi via `pyomo`, falling back to CBC when no commercial licence is available. Expected wall-clock runtime is below 60 seconds.

Genetic Algorithm with LP slave. For the trust-level formulation with $|\mathcal{I}|, |\mathcal{J}| \approx 150$, the exact MILP becomes expensive. We exploit the structural property that, for fixed binary $\mathbf{x}$, the residual problem in (b, z, u) is a linear programme. A master-slave decomposition uses a Genetic Algorithm over the $|\mathcal{J}|$ -dimensional binary vector $\mathbf{x}$, with one LP solve per fitness evaluation. The configuration is: tournament selection ($k = 3$), uniform crossover ($p_c = 0.9$), bit-flip mutation ($p_m = 1/|\mathcal{J}|$), elitism (top 5), population 100, maximum 200 generations with early-stopping on stagnation, and a budget-repair operator that deactivates facilities with the lowest cost-effectiveness ratio when constraint (10) is violated. This decomposition is standard in capacitated facility-location heuristics.

NSGA-II for the multi-objective formulation. The Non-dominated Sorting Genetic Algorithm II [6] produces a Pareto front over three competing objectives evaluated simultaneously without scalarisation:

$$f_1(\mathbf{x}) = \sum_j F_j x_j + \sum_{j,h} g_j b_{j,h}^*(\mathbf{x}) \quad (\text{infrastructure cost}) \quad (15)$$

$$f_2(\mathbf{x}) = \sum_{i,h,s} \pi_s u_{i,h}^{s*}(\mathbf{x}) \quad (\text{expected unmet demand}) \quad (16)$$

$$f_3(\mathbf{x}) = \sum_{i,j,h,s} \pi_s c_{ij} z_{ij,h}^{s*}(\mathbf{x}) \quad (\text{transfer burden}) \quad (17)$$

where the starred quantities are LP-slave optima for the given chromosome $\mathbf{x}$. We use the `pymoo` library with binary random sampling, two-point crossover, and bit-flip mutation; population size 100, maximum 200 generations, identical p_c, p_m to the single-objective GA. Solution quality is reported via hypervolume relative to a fixed reference point, generational distance, and the cardinality of the non-dominated set.

Simulated Annealing comparator. A standard Simulated Annealing comparator uses geometric cooling ($T_0 = 100$, $\gamma = 0.95$, stopping at $T_{\text{stop}} = 0.01$) over single bit-flip neighbourhoods, with the same LP slave at every step.

Heuristic baselines. For completeness we include *population-proportional*, *demand-proportional*, and *greedy shortage-first* allocation heuristics. Each fixes $\mathbf{x}$ by a closed-form rule, then assigns (b, z, u) via the same LP slave, providing a lower performance bound on optimisation-based methods.

5 NHS England Case Study

5.1 Data and Spatial Resolution

The case study uses three publicly available data sources covering 1 August 2020 to 31 August 2022, the window during which NHS England published daily regional admissions and bed-occupancy data: NHS England COVID-19 Hospital Activity (daily admissions and mechanical-ventilation bed occupancy), NHS England Urgent and Emergency Care Bed Availability (adult critical-care capacity at acute-trust level, providing $C_{j,h}$), and ONS open geography and mid-year population estimates (region boundaries, centroids for travel-time computation, and population denominators p_i). The main forecasting and regional optimisation experiments use NHS England’s seven regions: North East and Yorkshire, North West, Midlands, East of England, London, South East, and South West. Trust-level experiments use a subset of acute trusts with reported critical-care capacity. Travel times T_{ij} are approximated by great-circle distances scaled by a 1.3 detour factor, calibrated against OSRM driving estimates on a sample of routes.

5.2 Train, Validation, and Test Splits

The data are split chronologically by epidemic wave, deliberately holding the most clinically distinct wave in the test set: training on 1 Aug 2020 – 31 May 2021 (Alpha and early vaccination), validation on 1 Jun – 30 Nov 2021 (Delta) for hyperparameter tuning and early stopping, and testing on 1 Dec 2021 – 31 Aug 2022 (Omicron and post-Omicron). Omicron is held out because its case-to-hospitalisation and hospitalisation-to-critical-care ratios differ markedly from earlier waves, providing a stress test of the framework’s generalisation.

5.3 Optimisation Parameters

The fixed activation cost F_j and marginal capacity cost g_j are reported in scaled units, allowing sensitivity analysis without committing to monetary figures that depend on commissioning context. The budget B is varied across three levels (low, medium, high) to expose the cost–shortage trade-off, the travel-time threshold τ over $\{60, 120, 240\}$ minutes, and equity and CVaR weights λ_2, λ_3 via sensitivity sweeps. Default loss weights are $\lambda_1 = 10^3$, $\lambda_2 = 10^2$, $\lambda_3 = 10^1$, calibrated to bring the four objective components to comparable magnitudes. All experiments use fixed random seeds; results are reported as means over five seeds where applicable. Code, configurations, and the cleaned regional dataset are available at the repository linked in the acknowledgements.

Table 1. Forecasting accuracy on NHS regional mechanical-ventilation bed occupancy (lower RMSE is better) and underestimation rate at the 28-day horizon. Mean over 5 seeds; standard deviations in supplementary material. The bottom block ablates the composite training loss. **[TODO: fill from E1]**

Model	RMSE by horizon (days)				Underestim. rate (%)
	7	14	21	28	
Persistence	—	—	—	—	—
ARIMA	—	—	—	—	—
XGBoost (lag features)	—	—	—	—	—
LSTM	—	—	—	—	—
PINN-SEIRD + GRU	—	—	—	—	—
Proposed (decision-aware)	—	—	—	—	—
No PINN regularisation	—	—	—	—	—
No underestimation penalty	—	—	—	—	—
No smoothness	—	—	—	—	—

6 Results

We report results for six experiments: forecasting accuracy (E1), regional allocation (E2), trust-level metaheuristic comparison (E3), Pareto-front analysis (E4), robustness to forecast quality (E5), and sensitivity analysis (E6). All results use the test split (1 Dec 2021 – 31 Aug 2022).

6.1 Forecasting Accuracy (E1)

Table 1 reports root mean squared error (RMSE) by horizon for the proposed PINN-GRU with decision-aware loss against statistical, machine-learning, and recurrent baselines on mechanical-ventilation bed occupancy. The proposed model trades a small amount of root-mean-squared accuracy for a substantially lower underestimation rate, consistent with the asymmetric design of $\mathcal{L}_{\text{under}}$ in Equation (4).

6.2 Regional Allocation (E2)

Table 2 compares the deterministic and robust MILP solutions to allocation baselines on the seven-region NHS England case study, fixing the medium budget level and 120-minute travel-time threshold. The robust MILP achieves the lowest worst-case shortage, while deterministic MILP and demand-proportional allocation tie on expected unmet demand under the median forecast.

6.3 Metaheuristic Comparison (E3)

On a trust-level subset where MILP remains tractable, the Genetic Algorithm, NSGA-II (best of front via weighted-sum scalarisation), and Simulated Annealing

Table 2. Allocation comparison on the seven-region NHS England case study. [TODO: fill from E2]

Policy	Unmet (beds)	Coverage (%)	Travel (min/pat.)	Worst-case shortage
Current capacity only	—	—	—	—
Population-proportional	—	—	—	—
Demand-proportional	—	—	—	—
Greedy shortage-first	—	—	—	—
Deterministic MILP	—	—	—	—
Robust MILP (this work)	—	—	—	—

all achieve solution quality within [TODO: X]% of the MILP optimum at [TODO: Y] $\times$ lower runtime, supporting their use at full trust scale where MILP becomes expensive.

6.4 Pareto-Front Analysis, Robustness, and Sensitivity (E4–E6)

The NSGA-II Pareto front in the three-objective space (cost, expected unmet demand, transfer burden) is reported at trust-level scale, with sensitivity analysis of unmet demand against budget B and travel-time threshold τ . Robustness to forecast quality (E5) bracketed: replacing the proposed forecaster with a historical-median baseline raises unmet demand by [TODO: Z]%, ARIMA by [TODO: Y]%, while ground-truth oracle forecasts reduce it by [TODO: W]%.

7 Discussion and Limitations

Equity versus efficiency. The Pareto-front analysis highlights a recurring tension in allocation policy: the lowest-cost configuration concentrates surge capacity in the largest-demand regions, but achieves higher maximum-regional-shortage ratios than the equity-aware variant. The equity-linearised constraint (Equations (12) and (13)) shifts the front toward more uniform per-capita coverage at a modest cost premium. This trade-off is operationally relevant: equity-blind allocation can leave outlying regions exposed, while strict equity weighting can under-utilise concentrated capacity. The framework gives planners a principled way to elicit and explore that trade-off rather than burying it in a single objective.

Forecast quality, decision quality, and asymmetry. The forecast-value experiment shows a roughly [TODO: Z]% gap in unmet demand between the proposed forecaster and a historical-median baseline, and a [TODO: W]% gap to oracle forecasts. The asymmetric underestimation penalty (Equation (4)) is doing meaningful work: ablating it raises the underestimation rate by [TODO: X] percentage points (Table 1) without proportionally improving root mean squared error. Better accuracy on the wrong side of the forecast distribution still produces

shortages: this is the operational point of decision-aware training. At regional scale, deterministic and robust MILP solve exactly in seconds; at trust scale the LP-slave metaheuristics match within **[TODO: X]**% of MILP at **[TODO: Y]**× lower runtime, extending applicability to operational instance sizes.

Limitations. Costs F_j, g_j, c_{ij} are reported in scaled units; operational use would require commissioning-context calibration. The SEIRD model assumes well-mixed regional populations. Travel times are approximated by detour-scaled great-circle distances; an OSRM-derived matrix may shift particular assignments without changing qualitative conclusions. The framework is configured for NHS England structures; transfer to other health systems would require re-specifying demand units, capacity categories, and equity proxies, though the optimisation core is general. Natural extensions include graph-based forecasting backbones, a rolling-horizon online formulation, and a coupled budget-allocation upper level.

8 Conclusion

We presented a forecast-to-decision optimisation framework for NHS critical-care surge-capacity planning that links per-region physics-informed epidemic forecasting (with a refined $SEI_a I_s HCRD$ compartmental structure and a decision-aware composite loss) to a multi-objective robust mixed-integer programme, solved exactly at regional scale and via GA, NSGA-II, and Simulated Annealing at trust scale. By aligning forecast errors with downstream operational cost and exposing equity–efficiency trade-offs explicitly, the framework supports transparent and reproducible critical-care surge planning on open NHS data.

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